Responses of Infant Titi Monkeys, *Callicebus moloch*, to Removal of One or Both Parents: Evidence for Paternal Attachment

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The response to parental separation in infant titi monkeys was evaluated. Separation from the mother for 1 hr did not elicit an adrenocortical response from the infant unless the father was also removed. Separation from the father elicited a significant elevation in adrenocortical activity even when the mother remained with the infant during the separation period. Infants showed the highest cortisol levels and vocalization rates when both parents were removed and the infant remained alone in the living cage for 1 hr. As in previous research, infants maintained higher levels of contact with the father than with the mother. The results indicate that in this monogamous New World primate, the father is the primary attachment figure for the developing infant. © 1995 John Wiley & Sons, Inc.

The long period of infant dependency characteristic of mammals is often associated with the development of a filial bond or emotional attachment with the mother. Such

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a bond is manifested in the infant's selective preference for its mother, display of agitation or distress upon maternal separation, and the ability of the mother to attenuate the infant's response to potentially stressful circumstances (Ainsworth, 1972; Bowlby, 1973). For most primate species, the filial bond is reciprocated by an intense attachment of mother to infant (e.g., Mendoza, Coe, Smotherman, Kaplan, & Levine, 1980).

In the absence of the mother, infants will readily redirect their filial attachment to an inanimate or animate mother substitute (Mason, 1971; Mason & Kenney, 1974). As the classic Wisconsin experiments demonstrated with rhesus monkeys, a surrogate mother need not be a source of food, transport, or protection to become the object of a strong filial bond. Instead, it appears that tactile qualities of the attachment figure (i.e., contact comfort) predominate (Harlow & Zimmermann, 1959).

A monogamous South American primate, the titi monkey (*Callicebus moloch*), is particularly interesting in this regard because it represents a rare natural distinction between the parent providing food and the parent contacting the infant most often. The titi monkey infant is nutritionally dependent on the mother for the first 8 to 12 weeks of life; yet, the infant spends 70–90% of its time being transported by the father (Fragaszy, Schwarz, & Shimosaka, 1982; Jantschke, 1992; Meritt, 1980). Moreover, young titi monkeys continue to exhibit a selective preference for their fathers when they are no longer dependent on either parent for food or transport (Mendoza & Mason, 1986b). To date, measures of parental attachment in infant titi monkeys other than preference have not been evaluated.

In contrast to squirrel monkeys in which mothers clearly show a strong adrenocortical response to separation from their infant (Mendoza et al., 1980), studies of attachment in adult titi monkeys indicate that neither parent responds to separation from their dependent offspring (Mendoza & Mason, 1986b). Brief separation from their mate, however, leads to increased heart rate, cortisol levels, and behavioral distress (Cubicciotti & Mason, 1975; Mendoza & Mason, 1986a/1986b). The presence of the mate also decreases adrenocortical responsiveness to novelty or disturbance (Hennessy, Mendoza, Mason, & Moberg, 1995). Finally, in preference-testing situations, both males and females are more strongly attracted to their mate than to their offspring or to unfamiliar adults of either sex (Mason, 1975; Mendoza & Mason, 1986b). Thus, it appears that the relationship between adult titi monkeys resembles the filial bond in other species, but surprisingly, neither parent shows evidence of a comparable bond with their infant (Mendoza & Mason, 1986b).

The present experiment investigated the filial attachment of titi monkeys by examining the infant's behavioral and adrenocortical responses to separation from its parents. Of interest was whether infant titis would respond to separation from their parents as parents respond to separation from each other—with a robust behavioral and adrenocortical reaction, or as parents respond to separation from their infants—with little or no reaction. Further, if infants did respond to separation, would they respond more strongly to separation from their mother, as would be expected based on studies with other primate species or would they respond more strongly to separation from their father, as the preference data predict?

Methods

Subjects and Living Arrangements

The subjects for the experiment were 4 female and 2 male titi monkey infants (*Callicebus moloch*) living with their parents in six family groups. At the beginning of

data collection, the infants were between 3 and 5 months of age. Five family groups also contained two additional offspring, which were between 1 and 3 years older than the subjects. In one family group, the subject had no siblings. Three fathers and 3 mothers were wild-born and imported as adults; the other parents were colony-born and raised in family groups. Parents had been living together for at least 3 years before the study began. All animals except the infants were trained to enter small transport cages; infants generally entered transport cages with a parent.

Living cages measured 1.22 × 1.22 × 2.13 m and were situated such that each family group was visually isolated from others, but auditory and olfactory interactions were possible. All cages were equipped with four horizontal perches, arranged in stepwise fashion. Lighting was maintained on a 12 : 12 hr light : dark cycle with lights on at 0600 hr. Maintenance procedures followed standard laboratory protocol including daily cleaning and twice daily feeding at 0800 and 1330 hr (commercial monkey chow supplemented with fresh fruits and vegetables).

Procedure

All subjects were tested once in each of five experimental conditions: (a) base, (b) disturbance-control (infant-with-male-and-female), (c) infant-with-male, (d) infant-with-female, and (e) infant-alone.

In the base condition, the infant was captured, manually restrained, and a blood sample (.5 ml) collected as soon as possible for subsequent assay of plasma cortisol levels. In conditions 2 through 5, all family members were caught and placed in separate transport cages, and the infant (plus the mother and/or father as appropriate) was returned to the home cage as quickly as possible. The infant was always the first animal to be returned to the cage and was placed on the top perch; the parents were then released through a small entrance door at the bottom of the cage. Nonparticipating family members were removed from the colony room and housed at a remote location until the test session was complete. After 1 hr, the infant was recaptured and a blood sample was collected.

The order in which conditions were imposed was approximately balanced so that all subjects were exposed to the test conditions in a distinct sequence and at least 1 subject was tested in each condition during each week of testing. At least 5 nontest days intervened between successive conditions for a given subject; testing was completed with a 5-week period. Only one family group was tested in an experimental condition per day. All testing occurred between 1600 and 1730 hr.

Behavioral Observation

At the beginning of testing before any disturbance to the animals, the identity of the family member carrying the infant, if any, was recorded. During experimental conditions, behavioral observations were conducted using the infant as the focal subject. (These observations were not performed during the base condition.) The infant's behavior and interactions with its parents (when present) were recorded for the first 15 min following return of the infant (and parents), and again starting at 45 min following return.

Behaviors were recorded using a time-ruled checklist of predefined behavioral categories. At 15-s intervals, indicated by audible signals from an interval timer, dorsal contact, other contact, tail-twining, and proximity were recorded (on-the-signal). Locomotion, grooming, eating/drinking, rejection, retrieval, and grabbing/hitting/biting were recorded once if they occurred during a particular 15-s period (1–0). In addition to the

checklist behaviors, the total number of vocalizations emitted by the infant and its parent(s) was counted during the 30-min observation period. Instances of eating/drinking, retrieval, and grabbing/hitting/biting occurred too infrequently for data analysis, and will not be discussed further.

All behavioral data were recorded by the same trained observer. Reliability estimates obtained prior to data collection exceeded 85% agreement for all behavioral categories. The criteria used for recording the behaviors were as follows:

1. dorsal contact: infant on parent's back and weight of infant supported by the adult. (Note: Infant titi monkeys are rarely carried ventrally.)

2. other contact: any contact between infant and parent(s) in which the infant was supporting its own weight. This usually involves side-by-side contact.

3. tail-twining: A species-specific form of affiliative contact that occurs when animals are sitting side-by-side and one animal's tail is wrapped around the tail of another. All 3 animals could be tail-twining simultaneously.

4. proximity: within arm's reach of an adult. If the entire triad was positioned closely together, but one adult was located between the infant and the other adult, the infant was considered to be in proximity to the closest parent only.

5. locomotion: independent movement of at least one body length. This was not scored if the infant was being carried.

6. grooming: An individual gently picks through the hair of another individual.

7. rejection: attempts by the adult to remove the infant from the dorsal surface. This is often accompanied by biting the infant's hands or feet, or scraping the infant against the cage. It is also scored when the adult actively resists the infant's attempts to climb onto its back. This measure included unsuccessful rejection attempts, in which the infant remains on the adult's back or successfully climbs onto the back despite resistance.

8. vocalization: any oral sound emitted by the infant or its parents during the observation period.

The procedures for collecting vocalization data were somewhat more complex than those for other behaviors. The majority of vocalizations that are produced during separation conditions are high-pitched distress vocalizations, also called isolation peeps. Because titi monkeys can produce these vocalizations with their mouths closed, and because adults will vocalize when separated from their mates, it is not possible to distinguish infant vocalizations from adult vocalizations. Moreover, it is not always possible to reliably distinguish these vocalizations from other titi vocalizations in type or function. Therefore, the cumulative number of vocalizations emitted by all individuals remaining in the home cage was recorded. A further complication arose from our decision to observe the subjects in the colony room rather than in a separate room isolated from the influence of other monkeys. During some 15-s intervals, loud calls produced by other monkeys in the room made it impossible to determine whether any vocalizations were produced by the animals being observed. When this occurred, "room noise" was marked on the behavioral checklist and no vocalizations were recorded for that 15-s interval. For analysis, vocalizations per minute were calculated for each experimental condition, and included only those 15-s intervals in which room noise did not occur. Room noise occurred in fewer than 10% of the intervals overall and was not different among the various experimental conditions.

Blood Sampling and Endocrine Evaluations

Blood samples were taken by femoral venipuncture while the monkey was manually restrained. The average time between entering the cage and collection of the sample

was 2.61 min (range = 1.20 to 4.57 min). Based on studies with laboratory rodents and squirrel monkeys, this was rapid enough to ensure that cortisol levels in the samples collected were not influenced by the blood-sampling procedures (Davidson, Jones, & Levine, 1968; Lyons, Mendoza, & Mason, 1994). Samples were centrifuged, and the plasma fraction was extracted and frozen (−20°C) until assayed. Evaluations of plasma cortisol concentraitons were made using commercially prepared kits (Diagnostics Products Corporation). Samples were diluted 1 : 4 prior to assay, and all samples were run in a single assay.

Data Analysis

Infant cortisol levels were analyzed using a one-way repeated measures analysis of variance. Typically, the behavioral measures were either frequently expressed (near or at 100%) or infrequently expressed (near or at 0). In order to avoid ceiling or floor influences in the analysis, all behavioral data were analyzed nonparamerically using Friedman analyses of variance or Wilcoxon signed-ranks tests. The small sample size prevented adjusting p values for the number of tests performed. Preliminary analysis revealed few behavioral changes from the first to the last 15 min of observation and, hence, are not reported. Post-hoc comparisons were performed, where appropriate, using Newman-Keuls (cortisol) or Wilcoxon tests (behavior).

Results

During pretest observations, infants were found to be in dorsal contact on 19 of 30 occasions. Infants were never observed on their mother or siblings and all were carried by their father on at least one occasion. Thus, when infants were not found on their fathers, they were on their own.

In the disturbance-control condition, during which both parents were present, infants spent most of their time in proximity to at least one parent, M = 86.3% of observations; r = 65.8–100.0%. In this condition, infants were in proximity to their fathers significantly more often than to their mothers, M = 76.1% and 32.8% of the intervals, respectively; $p < .05$. When in proximity, infants tended to phsyically contact (dorsal contact, other contact, tail-twine) their fathers more often than their mothers (contact with fathers: 64.9% of observations; contact with mothers: 13.2% of observations; $p < .05$). Much of the contact between infants and fathers was dorsal contact, but this type of contact was rarely observed with the mothers (dorsal contact with fathers: 39.0% of observations; dorsal contact with mothers: 1.0%; $p < .05$).

In the infant-with-male condition, infants maintained high levels of proximity to and contact with their fathers (proximity: 95.8%; contact: 87.9%; dorsal contact: 52.4%). During the infant-with-female condition, infants spent more time in proximity to and contact with their mothers (proximity: 67.6%, $p = .07$; contact: 57.1%, $p < .05$) than they did in the disturbance-control condition. Dorsal contact remained at low levels when only the mother was present (6.9%). In spite of the tendency for infants to spend more time with their mothers during the infant-with-female condition, infants still spent significantly more time with their fathers than with their mothers when only one parent was available (proximity, $p < .05$; contact, $p < .05$; dorsal contact, $p < .05$).

Only one rejection was observed during the disturbance-control condition, when both parents were present. Maternal rejection increased during the infant-with-female condition, $p < .05$, during which the majority of rejections (34 of 41) were observed.

Table 1
Mean Number and Range of Vocalizations Detected During the First and Last 15 Min of Each 1-hr Test Period

Condition	Mean	Range
Disturbance-Control	23.0	4.8–49.8
Infant-with-Father	78.2	7.8–249.6
Infant-with-Mother	214.1	34.2–727.2
Alone	340.8	37.2–420.0

Not all infant–female interactions were negative. Although tail-twine was infrequent, the infants were observed tail-twining with their mothers (55 occasions) as well as with their fathers (113 occasions). Grooming also occurred at least as often between mother and infant (36 instances) as between father and infant (17 instances). Differences between males and females in tail-twining and grooming with their infants were not significant.

Infant activity levels were marginally influenced by experimental conditions. Infants tended to be most active during the infant-alone condition (Locomotion was scored during 52.3% of the intervals.), somewhat less active in the infant-with-female condition (30.2%) and disturbance-control condition (23.6%), and least active in the infant-with-male condition (12.3%). The difference among conditions in locomotion scores approached significance, $p = .072$.

Vocalization significantly differentiated test conditions, $p < .05$; see Table 1. Post-hoc analyses indicated that the number of vocalizations emitted by the infant and parents (when present) was significantly higher in the infant-alone condition than in the disturbance-control condition and infant-with-male condition, $ps < .05$. Vocalizations also occurred significantly more often in the infant-with-female condition than in the disturbance-control condition, $p < .05$. No other comparisons of conditions were significant.

Analysis of cortisol levels indicated that the experimental conditions significantly altered adrenocortical activity, $F(4,20) = 25.18$, $p < 0.001$; see Figure 1. Infants did not respond to removal of just the mother; cortisol levels when only the mother was removed (infant-with-male condition) did not differ from the disturbance-control or base conditions in which both parents were present. Infants did respond to removal of just the father. Cortisol levels when only the father was removed (infant-with-female condition) were significantly higher than in any of the three conditions in which the father was present (base, disturbance-control, infant-with-male, $ps < .01$). The highest cortisol levels were found in the infant-alone condition; these levels significantly differed from each of the other experimental conditions, $ps < .01$. Each of the 6 subjects showed the same general pattern of response; cortisol levels were always higher in the infant-alone condition than in the infant-with-female condition, and always higher in the infant-with-female condition than in the infant-with-male condition. In sum, cortisol levels show that the infants responded to removal of the father from the living cage, even when the mother was present; whereas infants did not respond to removal of the mother when the father was present. This does not imply that the mother was without effect, however, for the infants responded most vigorously when left entirely alone in the living cage.

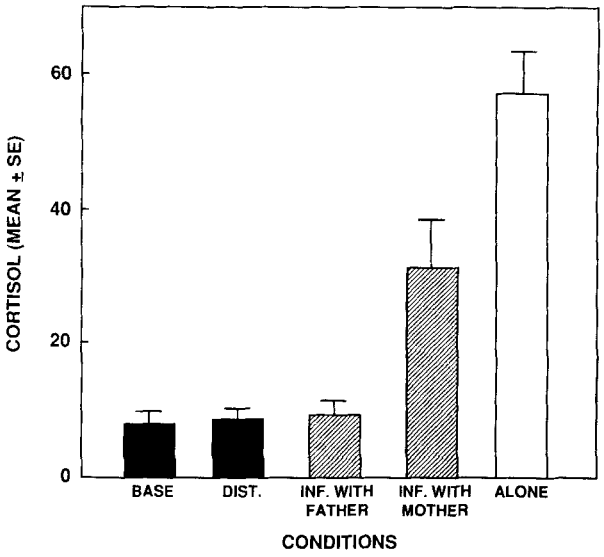


Fig. 1. Mean (±*SE*) plasma cortisol levels of infants following no disturbance (base), or 1 hr following disturbance only (dist), removal of the mother (infant-with-father), removal of the father (infant-with-mother), or removal of both parents (alone).

Discussion

The behavioral and physiological data clearly indicate that infant titis do respond to separation from their parents. When alone, infants exhibited the highest levels of cortisol and the highest rate of vocalizations, measures which generally indicate stress and agitation. Although it was not possible to count the number of infant vocalizations without also including those produced by the parents, the main result that vocalizations were infrequent when both parents were present and most frequent when the infant was alone is unambiguous.

It is also clear from our data that infants did not respond to separation from their mothers and their fathers equivalently. Separation from the father was a more potent stressor than separation from the mother. Removal of the mother did not lead to increased cortisol levels unless the father was also removed; removal of the father elicited an increase in plasma cortisol levels even when the mother remained with the infant during the separation period.

Capture and handling procedures have been shown to increase adrenocortical activity in adult *Callicebus* (Mendoza, 1991). We presume that in the present study, catching and returning the infant was a stimulus sufficient to increase adrenocortical activity as well. In this context, then, the father seems to completely eliminate the infant's adrenocortical response to experimental disturbance, whereas the mother does not seem capable of doing so.

Whether increased secretion of cortisol or other hormones of the hypothalamic–pituitary–adrenal axis in any way affects the attachment process or filial behavior of titi monkeys is unknown. However, results with other species suggest that increased release of corticotropin-releasing factor may play a role in the behavioral responses to separation. This neuropeptide not only stimulates pituitary release of the corticotroph, but

also orchestrates various other physiological and behavioral responses to stressors (Dunn & Berridge, 1990), and may modulate or eventually terminate vocalizing and other active behavioral responses to separation (Harvey & Hennessy, in press; Hennessy, Long, Nigh, Williams, & Nolan, in press; Kalin, Shelton, & Barksdale, 1989).

Our behavioral findings replicate previous results indicating that infants, when given the opportunity to be with either parent, prefer to be with their fathers (Mendoza & Mason, 1986b). Even when only one parent was present (infant-with-male and infant-with-female conditions), the infant tended to be in proximity and contact with their fathers more than their mothers. It follows from the patterns of parent–infant contact and the infant's adrenocortical and behavioral responses to separation that the father is the primary filial attachment figure.

This work appears to be the first demonstration in primates of a physiological stress response to brief separation from the father and contrasts with studies of other species in which an infant's response to separation had been observed only when the mother was removed (e.g., Coe, Mendoza, Smotherman, & Levine, 1978; Kaufman & Rosenblum, 1969; Reite, Short, Seiler, & Pauley, 1981; Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978; Vogt & Levine, 1980). We do not know from our findings whether the mother represents a special or distinct social category for the titi monkey infant. That is, it is possible that removal of any familiar social companion would have produced a cortisol elevation comparable to that observed when the mother was removed.

The differential responses displayed by infants to separation from their mothers and fathers could, at least partially, be due to the parental reaction during the separation period. A study by Hinde and Davies (1972) found that the amount of distress behavior displayed by rhesus infants when they were with their mothers depended on how perturbed the mothers were at the time. In our study, the mothers did appear more agitated than the fathers when they were alone with the infant. However, some of the agitation appeared to be due to the mother's attempts to avoid dorsal contact with her infant and, hence, provides further evidence for a weaker bond between mother and infant titi monkeys.

The major implications of this research concern the relationship between parental care, parental contact, and filial attachment. Titi monkeys are one of the few mammalian species in which the infant spends more time with its father than with its mother even though the mother is necessarily the source of food for the first 2–3 months of life. The results from this study indicate that the father, not the mother, is the primary attachment figure for the infant. Thus, the determining factor in the development of attachment in the infant titi is not the source of nourishment, but rather the source of contact.

Notes

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